

AI Engineering Manager - Database Design Specification

Collections Overview

Collection	Purpose
members	Stores single profiles, skills, and platform details for all active engineers and administrative roles.
teams	Groups developers together into their internal technical departments for quick tracking and reporting.
standups	Logs each daily batch update session, tracking structural workflow links and AI processing states.
standupMessages	Temporarily saves the raw text updates typed or spoken by each person before being parsed.
tasks	Acts as the core dashboard view to track active work summaries, priority levels, dependencies, and execution metrics.
activities / taskUpdates	Acts like a permanent diary that automatically records a snapshot of every single change made to a task, including collaborative comments.
notifications	Queues outbox logs for automated Slack alerts or reminders to ensure successful message delivery.
dependencies	Explicitly charts cross-task blocker preconditions and prereq mappings across squads.

members

Purpose: Stores profile information like names, roles, and platform details for every team member.

Why Created: To avoid repeating member profiles across separate tasks or updates, keeping data clean and centralized.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
name	String	Yes	Employee display name
email	String	Yes	Official workspace email address
slackUserId	String	No	Slack provider identification token
role	Enum / String	Yes	Developer/QA/ScrumMaster/CTO/Admin role parameters
skills	Array	No	Technical capability tags and skills profiling
teamId	ObjectId	No	Reference to parent engineering team tracking block
isActive	Boolean	Yes	Soft delete status flag
createdAt	Date	Yes	Timestamp
updatedAt	Date	Yes	Timestamp

teams

Purpose: Models logical company squads and structural organizational technical divisions.

Why Created: Allows management interfaces to filter views and analyze operational progress reports by department.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
name	String	Yes	Structural squad/department name

standups

Purpose: Keeps track of overall processing states for individual daily stand-up meeting submission entries. It specifies whether the submission node was hand-pasted, imported from Slack, or populated via API logs.

Why Created: To maintain an organized overview of daily logs and ensure the AI pipeline processes the inputs completely.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
submittedBy	ObjectId	Yes	Reference to submitter profile record
source	Enum	No	Manual/Slack/API variant log tag
parsingStatus	Enum	Yes	Processing lifecycle: Pending, Processing, Completed, Failed
message	String	No	Raw baseline incoming text feed
parsed	Boolean	No	Pipeline execution true/false toggle
createdAt	Date	Yes	Submission intake milestone timestamp

standupMessages

Purpose: Stores raw text segments or conversational transcripts mapping explicitly back to single speakers.

Why Created: To safely hold original textual descriptions exactly as submitted, allowing clean tracking or individual re-parsing.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
standupId	ObjectId	Yes	Reference to structural parent batch tracking record
memberId	ObjectId	Yes	Reference to individual active speaker record
rawMessage	String	Yes	Original status message typed/spoken by developer
parsed	Boolean	Yes	Parser handling state tracking variable

tasks

Purpose: Manages the current status parameters, descriptions, and priority levels of active work units. The description logs whether the entry was extracted by AI routines or manually drafted by hand.

Why Created: To provide a high-speed querying surface for dashboard items, tracking status variables without performance latency.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
memberId / ownerId	ObjectId	Yes	Reference to assigned engineer owner record
standupId	ObjectId	Yes	Reference to source intake origin record
title	String	Yes	Short semantic summary label of action or features
description	String	No	Granular tactical description context details
status	Enum / String	Yes	Progress execution state: PROCESSING, COMPLETED, BLOCKED
workflowStage	Enum / String	Yes	Pipeline role tracking: DEVELOPMENT, QA, REVIEW, PRODUCTION
priority	Enum / String	Yes	Low, Medium, High, Critical tracking markers
deadline	Date	No	Target completion timeline parameters
createdAt	Date	Yes	Timestamp
updatedAt	Date	Yes	Timestamp

activities / taskUpdates

Purpose: It acts like a permanent diary that automatically records a snapshot of every single change made to a task.

Why Created: To show a step-by-step history timeline on your dashboard without cluttering or slowing down your main task list.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
taskId	ObjectId	Yes	Reference tracing back to target parent operational record
actorType	Enum	No	USER/AI_AGENT/SYSTEM origin identifier variables
actorId / updatedBy	ObjectId / String	No	Reference tracing modification source trigger identity
activityType	Enum	No	STATUS_CHANGE, COMMENT, NOTIFICATION, FOLLOWUP tracking states
previousStatus	String	No	Historical trace state value prior to shift
currentStatus	String	No	Successive transition logging state attribute
oldValue	Mixed	No	Generic catch-all historical value prior to operation pass
newValue	Mixed	No	Generic successive operational update property attribute
message / updateMessage	String	No	Contextual semantic textual tracking descriptions or summaries (e.g., houses the text string when activityType = COMMENT)
createdAt / updatedAt	Date	Yes	Immutable ledger entry milestone timeframe

notifications

Purpose: Maintains an outbox queue tracking the delivery execution parameters of system notifications.

Why Created: To track automated communication workflows and handle delivery validation across channels smoothly.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
taskId	ObjectId	No	Optional linking context pointer parameter
recipientId	ObjectId	Yes	Reference tracking destination target worker profile
senderType	Enum	Yes	AI/SYSTEM/USER configuration properties
channel	Enum	Yes	Slack, Email, App delivery selection identifiers
status	Enum	Yes	Intake delivery monitoring: Queued, Sent, Delivered, Read, Failed
createdAt	Date	Yes	Timestamp

dependencies

Purpose: Explicitly maps blocking relationships and pre-requisite chains between distinct tasks across squads.

Why Created: To build clear operational networks so the AI agent instantly recognizes bottleneck configurations.

Field	Type	Required	Notes / Descriptions
_id	ObjectId	Yes	Primary Key
taskId	ObjectId	Yes	Reference mapping out the dependent child action record
dependsOnTaskId	ObjectId	Yes	Reference pinpointing blocking source parent action record
dependencyType	Enum	No	Structural properties classification (e.g., Blocks, Relates To)